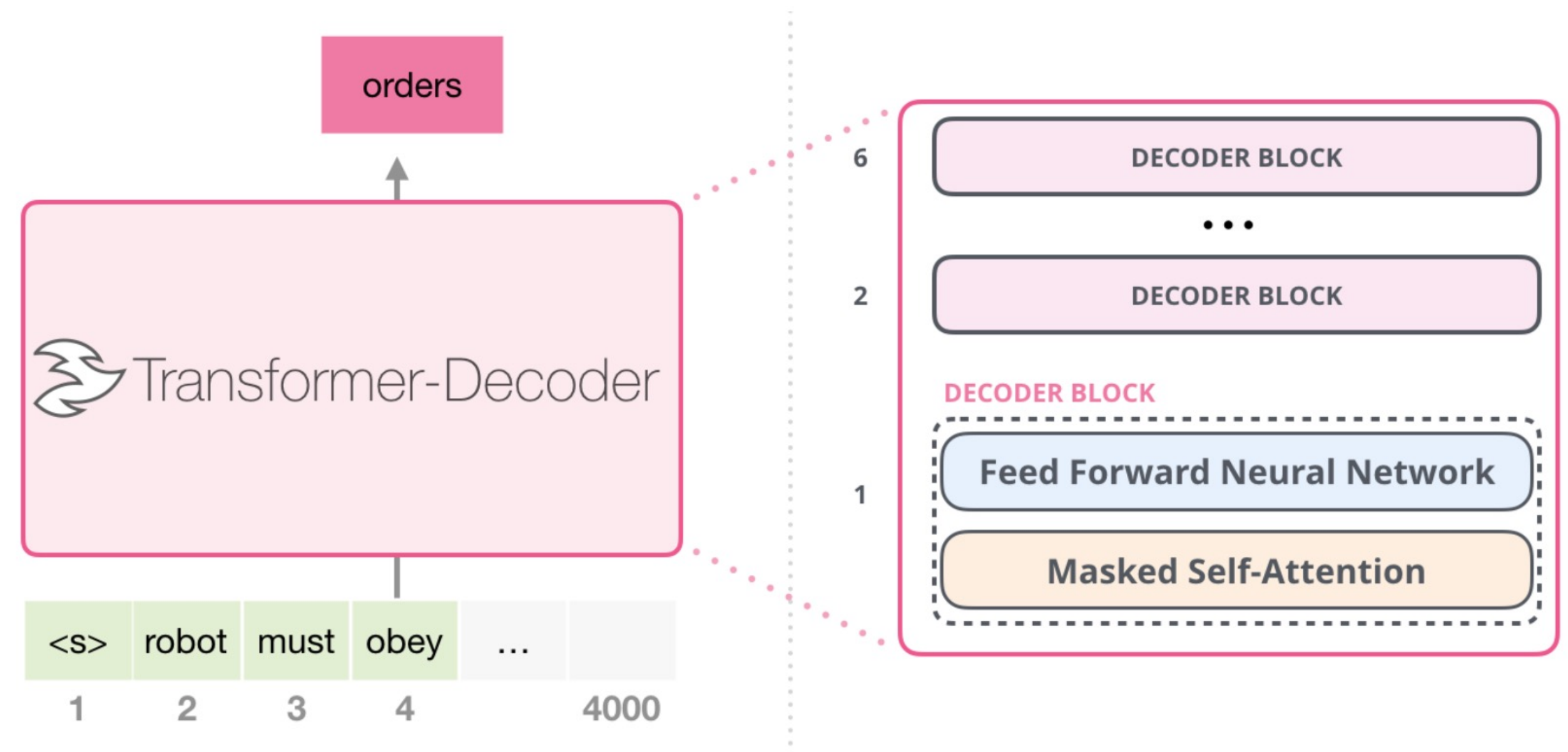


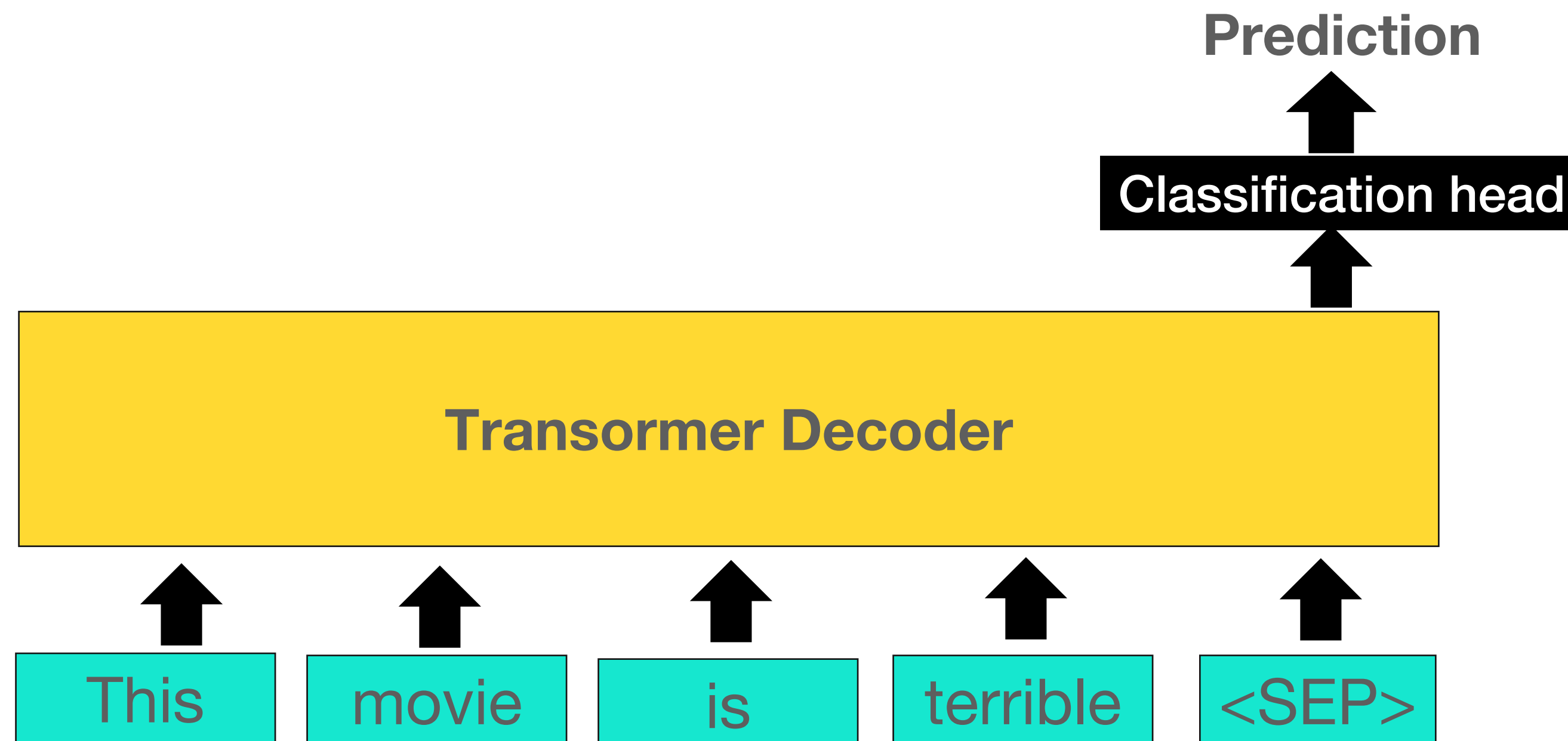
Decoder-only Models

- Another prominent alternative to Transformer-based language models are decoder-only models
 - Sometimes referred to as GPT (generative pretraining) models
- These models are left-to-right language models, consisting only of a Transformer decoder



Fine-tuning Decoder-only Models

- Decoder-only models can also be fine-tuned
 - Implementation details vary but the crux is the same – using the parameters of the pretrained model as the initialization for a prediction model built on top of it



Left-to-right vs. Masked Language Models

- There has been much debate over recent years as to the relative merits of (ltr) LMs and MLMs
 - LMs define a probability distribution over strings, while MLMs do not
 - MLMs take both sides of the context into account in their representations (LMs only have representations of prefixes)
 - LMs can be easily fine-tuned not only for prediction, but also for text generation (this property turns out to be crucial)